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RESEARCH ARTICLE

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Key Points:

- Optimal transport modeling allows for the solution of underdetermined source-pathway transport problems
- Spatial heterogeneity in the connectivity of sources and pathways shifts the dominant couplings of nitrate transport
- We provide a new tool for identifying source-pathway couplings that lead to high nutrient loading to streams

Supporting Information:

- Supporting Information S1

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Optimal Transport for Assessing Nitrate Source-Pathway Connectivity

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Abstract Excessive nitrate threatens a wide range of water resources, aquatic habitats, and sensitive infrastructure. Despite this problem, tracing a nutrient from its eventual fate back to its origin remains an elusive challenge due to heterogeneity in how nutrient sources and hydrologic pathways are connected. Typically, this problem is underdetermined (i.e., too many unknowns, not enough equations) and cannot be solved with existing methodologies. The theory of optimal transport allows for the solution of underdetermined systems, and here we construct a novel formulation for its use in water quality modeling. Our objective was to develop an optimal transport modeling framework—coupled to Bayesian source unmixing, loadograph pathway separation, and geospatial connectivity analysis—to apportion nitrate loading from three sources (soil, fertilizer, and manure) across three pathways (quick, intermediate, and slow), resulting in nine possible source-pathway couplings (soil-quick, soil-intermediate, ..., manure-slow). We apply this model to a 30 month elemental (NO_3^-) and isotopic ($\delta^{15}\text{N}$ and $\delta^{18}\text{O}$) nitrate data set from a karst watershed in Kentucky, USA. Modeling results indicate that—of the nine possible source-pathway couplings—nearly 60% of nitrate export is facilitated by just three: fertilizer-quick (16.4%), manure-intermediate (15.4%), and soil-slow (27.2%). Further, we reinforce the need to explicitly consider heterogeneity in source-pathway connectivity as homogeneous assumptions lead to erroneous inferences. The applicability of the model, its input requirements, and transferability to other sites is discussed. Lastly, we simulated two land management scenarios (field buffers and septic repair) and demonstrate how optimal transport can be used to test nutrient reduction strategies.

1. Introduction

Nitrate (NO_3^-) pollution resulting from anthropogenic activity is one of the most common freshwater contaminants (Wang et al., 2016) and is linked to harmful algae blooms (Watson et al., 2015), increased water treatment costs (Almuhtaram et al., 2018), and reduced recreational water use (Brooks et al., 2016). Worsening water quality seems likely with modeling projections showing increases in global NO_3^- concentrations over the coming decades (Ascott et al., 2017; Wang et al., 2016). Developing tools to forecast water quality impairment is complicated by uncertainties regarding the source of nutrients, the linkage to pathways, and the shifting of hydrologic cycles as a result of climate change (Brooks et al., 2016; Miller et al., 2017). In addition to an understanding of nutrient sources and pathways, effective land management strategies require a deeper understanding of the connectivity of particularly vulnerable source-pathway couplings, which should be prioritized for remediation (Jarvie et al., 2017; Kumar et al., 2018).

Despite the urgency of this problem, tracing a nutrient from its eventual fate back to its origin remains an elusive challenge with uncertainties arising from heterogeneities in the connectivity of hydrologic pathways and nutrient sources (Figure 1). Pathways of nitrate broadly include quick, intermediate, and slow, and are analogous to runoff, interflow, and baseflow, respectively. Time scales of transport for quick pathways are on the order of hours to days while they range from weeks to months and months to years for intermediate and slow pathways, respectively (Benettin et al., 2020; Husic, Fox, Adams, Ford, et al., 2019; Mellander et al., 2012). Tools to quantify these pathways include hydrograph separation (Husic, Fox, Adams, Ford, et al., 2019; Lutz et al., 2018), numerical modeling (Husic, Fox, Adams, Ford, et al., 2019; Sullivan et al., 2019), and high-frequency sensing (Fenton et al., 2017; Mellander et al., 2012; Miller et al., 2017). Sources of nitrate can include atmospheric deposition, fertilizer, soil nitrogen, and animal

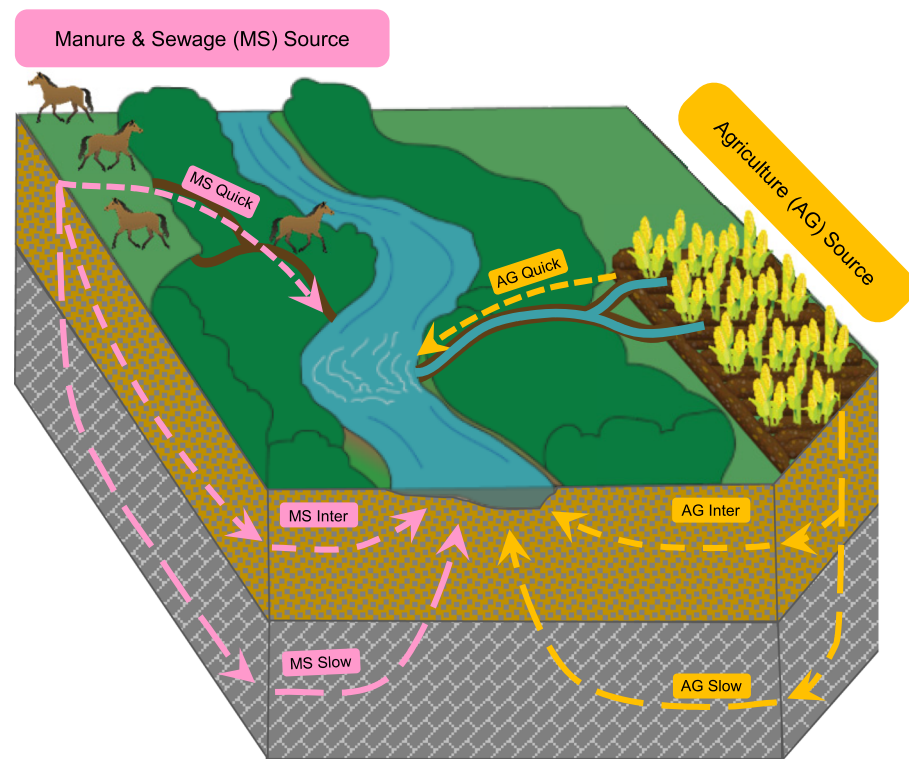


Figure 1. Conceptual model of multiple sources and pathways delivering nitrate in a mixed land use watershed. Sources can include manure and sewage, soil-derived nitrate, fertilizer, and atmospheric deposition. Pathways can include quick (e.g., drainage ditches), intermediate (e.g., vadose zone), and slow (e.g., deep groundwater) flow.

waste (Kendall et al., 2007; Xue et al., 2009) and are mostly commonly unmixed using a dual isotope ($\delta^{15}\text{N}_{\text{NO}_3}$ and $\delta^{18}\text{O}_{\text{NO}_3}$) mass balance approach (Hu et al., 2019; Li et al., 2019; Z. Zhang, Chen, et al., 2018; H. Zhang, Yang, et al., 2018; M. Zhang, Zhi, et al., 2018). However, the disjointed knowledge of hydrologic flow paths and source provenance alone is often inadequate to land managers as there can be substantial heterogeneity in distribution of source loadings across flow pathways (Botter et al., 2020; Musolff et al., 2017; Zhi et al., 2019). Further, the possible number of source-pathway couplings quickly increases as additional end-members are identified. For example, for three sources and three pathways there are nine possible couplings: soil-quick, soil-intermediate, ..., manure-slow. Typically, this problem is underdetermined (i.e., too many unknowns, not enough equations) and is intractable using existing tools and methodologies. New models capable of reconciling these limitations could improve how resources for load reduction are allocated given site-specific assessment of dominant source-pathway couplings.

Over the last few decades, the theory of optimal transport has allowed for the solution of underdetermined systems—where mass is redistributed from one location to another—by minimizing the cost of transport (Peyré & Cuturi, 2019; Villani, 2008). At its core, the theory answers the question “how can we move mass from point *a* to point *b* most efficiently?” (Villani, 2008). Optimal transport can be conceptualized as either the mapping of one continuous probability mass distribution onto another (i.e., the Monge formulation) or the transposition of discrete masses from one set of spatial locations to another (i.e., the Kantorovich problem) (Peyré & Cuturi, 2019). In either case, the global cost of transport is represented as the summation of the local cost of each individual mass transposition. Cost is typically represented by some weight factor, which is often the Euclidean distance but can be other measures of the relative work required for transport (Villani, 2008). This methodology has gained widespread use in a broad array of fields, including image processing, economics, and metapopulation dynamics (Nichols et al., 2017; Peyré & Cuturi, 2019; Santambrogio, 2015). Despite its utility to the analysis of complex data sets, it has not been conceptualized

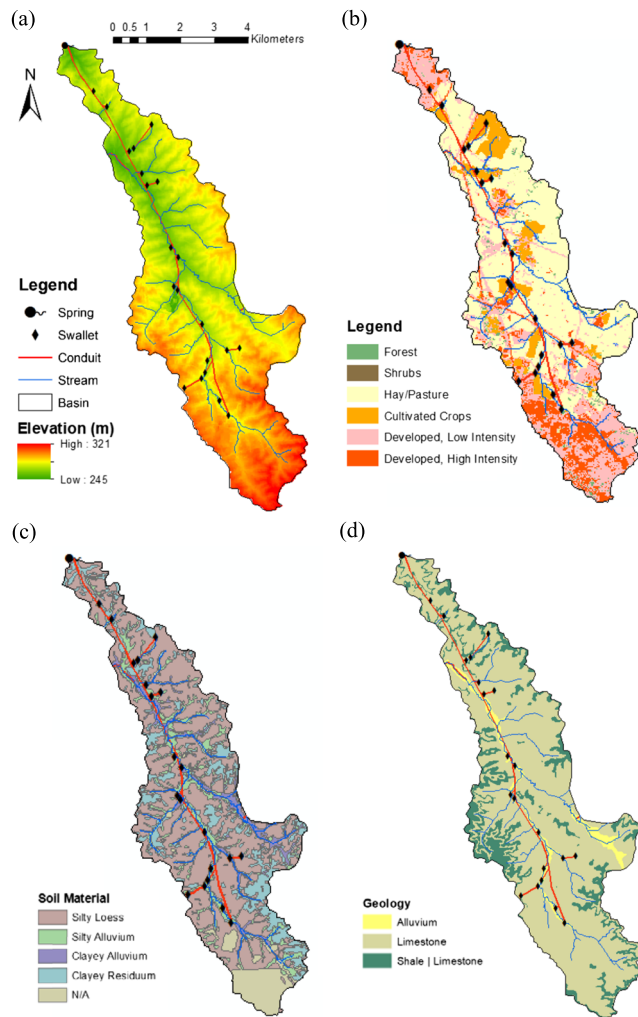


Figure 2. Maps of Royal Spring (a) streams, conduits, swallets, and topography; (b) land use and land cover; (c) soil composition; and (d) geology.

in the context of hydrologic science and engineering, fields in need of new, parsimonious modeling tools that integrate commonly collected data sets as inputs (Wellen et al., 2015).

The concept of redistributing mass (i.e., *connecting* the source and detachment of material to its transport and eventual fate) is tightly linked to the hydrologic and geomorphologic literature (Fryirs, 2013; Harvey et al., 2018; Lane et al., 2018; Mahoney et al., 2018). Connectivity is the mechanism that facilitates linkage between terrestrial and aquatic environments (Covino, 2017). Conversely, (dis)connectivity can be conceptualized as the sum of buffers, barriers, and blankets that impede transport across the landscape (Fryirs, 2013). The conveyance of matter across the landscape is heavily impacted by the heterogeneity in the spatial organization of likely source availability and preferential flow path connectivity (Mahoney et al., 2018; Musolff et al., 2017). In a homogeneous system, all material would equally be connected to all pathways, however this assumption often leads to erroneous process representation (Hartmann et al., 2017). The extent to which parcels of matter are connected to preferential hydrologic flow paths can be quantified using geospatial indices like the Index of Connectivity (*IC*) (Borselli et al., 2008; Cavalli et al., 2013; Crema & Cavalli, 2018) and the Topographic Wetness Index (*TWI*) (Buchanan et al., 2014; Güntner et al., 2004). While connectivity indices provide semiquantitative insights into the geospatial distribution of transportable matter and energy, their integration into numerical models is underdeveloped in the literature (Mahoney et al., 2018; Wohl et al., 2019).

In this paper, we propose a numerical coupling of connectivity theory with optimal transport theory to help resolve the intractability of the underdetermined nitrate source-pathway problem. Specifically, we utilize stable isotope end-members for source identification, lumped-modeling for pathway separation, and hydrologic connectivity for parameterizing transport cost. To the last point, we identify a unique solution to the transport matrix by minimizing watershed-scale (dis)connectivity. Minimizing (dis)connectivity has parallels to work and kinetic energy minimization, which underscores many physical and ecological processes (Nichols et al., 2017). The resulting optimal transport solution could provide a tool for implementing land management strategies that target dominant

source-pathway connections, and we explore this idea by simulating scenarios (e.g., installing buffer strips, repairing failing septic systems, and planting cover crops) to mitigate nitrate loading by human intervention best practices.

Overall, the objective of this study was to develop an optimal transport modeling framework for assessing source-pathway connectivity of nitrate in a human-disturbed watershed. We collected 30 months of elemental and isotopic nitrate data, performed geospatial connectivity analysis, extended an existing hydrologic and nitrogen model, and coupled that model to a novel optimal transport connectivity framework. Thereafter, we assessed the impact of heterogeneity and land management (e.g., repairing septic systems and planting cover crops) on redistributing mass in the source-pathway transport matrix. Lastly, we discuss the applicability and transferability of optimal transport modeling to other watersheds and consider the various methods by which data inputs for the model can be generated.

2. Materials and Methods

2.1. Study Site

The Royal Spring groundwater basin (58 km²; Figure 2a) is a mature karst watershed located in the Inner Bluegrass Region of Kentucky, USA, where the climate is temperate (mean annual temperature: 13.0 ± 0.7°C; mean annual precipitation: 1,170 ± 200 mm). Land use is primarily rural (60%) with urban

(40%) tributaries in the headwaters of Lexington, KY (Figure 2b). Soils in the watershed are moderately deep, well-drained (Figure 2c), and underlain by phosphatic limestone of the Middle Ordovician period (Figure 2d). The surface stream corridor runs dry over 80% of the year as it loses a significant amount of flow to seepage through cracks, fissures, and swallets. Swallets are defined as sinkholes located in the bed of streams and rivers, which connect surface and subsurface pathways. Subsurface flows are routed to a phreatic conduit that supplies the primary basin outlet (Royal Spring, 243 m above sea level) with an average perennial discharge of $0.67 \text{ m}^3 \text{ s}^{-1}$. Royal Spring serves as the raw municipal water source for 11,000 residents of Georgetown, KY (UKCAFE, 2011).

2.2. Nitrate Sources and Pathways

The United States Geological Survey (USGS) operates a v-notch weir at Royal Spring (USGS 03288110) where nitrate concentration and stable isotope composition samples were collected. In addition to the 30 new isotopic samples collected for the purpose of this study, 40 samples from prior work on storm-flow (Husic, Fox, Adams, Backus, et al., 2019) and low-flow (Husic et al., 2020) dynamics in Royal Spring were integrated into the analysis data set. All sets of samples followed the same collection, processing, and quality assurance procedures. Anion analysis was performed at the Kentucky Geological Survey laboratory, following EPA Method 300.0, on a Dionex ICS-3000 with a carbonate-bicarbonate eluent generator and Dionex AS4A analytical column. Lab ($n = 49$) and field ($n = 8$) duplicates had standard deviations of ± 0.02 and $\pm 0.07 \text{ mg N L}^{-1}$, respectively, and no lab, field, or equipment blanks registered above the method detection limit. Isotopic data was produced at the University of Arkansas Stable Isotope Laboratory with a Thermo Scientific GasBench II and Delta Plus IRMS using the bacterial denitrifier method (Casciotti et al., 2002). Isotope ratios are expressed with delta notation (δ) and are reported in units of per mil (‰) as the ratio of ^{15}N to ^{14}N in a sample versus a reference standard (e.g., $\delta^{15}\text{N} = [((^{15}\text{N}/^{14}\text{N})_{\text{sample}} / (^{15}\text{N}/^{14}\text{N})_{\text{air}}) - 1] \times 1,000$) (Kendall et al., 2007). The isotopic reference materials for NO_3^- were USGS32 ($\delta^{15}\text{N}_{\text{NO}_3} = +180\text{‰}$, $\delta^{18}\text{O}_{\text{NO}_3} = -27.9\text{‰}$), USGS34 ($\delta^{15}\text{N}_{\text{NO}_3} = -1.8\text{‰}$), and USGS35 ($\delta^{18}\text{O}_{\text{NO}_3} = +57.5\text{‰}$). Field duplicates ($n = 5$) of $\delta^{15}\text{N}_{\text{NO}_3}$ and $\delta^{18}\text{O}_{\text{NO}_3}$ had standard deviations of $\pm 0.28\text{‰}$ and $\pm 0.45\text{‰}$, respectively.

A calibrated, lumped hydrologic and nitrogen model of the Royal Spring watershed, published recently by the authors, provided inputs to the source-pathway framework developed in this study (supporting information Figure S1; Husic, Fox, Adams, Ford, et al., 2019). These inputs include a continuous estimate of spring NO_3^- concentration and a deconvolution of the spring NO_3^- load into quick, intermediate, and slow pathway end-members. These pathways represent sinkholes, fractures, and matrix pores, respectively. Further, extensive uncertainty analysis (i.e., over 30 million simulations) was performed on the numerical model, yielding a robust assessment of equifinality. In the previous Husic, Fox, Adams, Ford, et al. (2019) study, the historic model was calibrated and validated from October 2012 to October 2016. Here, we project the model forward from October 2016 to June 2019 and find good agreement between subsequently collected data and projected model results. The spatial and temporal resolutions of all data sets are presented in Table S1.

2.3. Optimal Transport Model Formulation

The optimal transport problem, in a discrete setting, aims to minimize an expected cost of moving mass from one location to another (Figure 3; Villani, 2008). To describe the starting mass distribution of sources prior to transport, we employed end-member mixing analysis (EMMA). The underlying equation for EMMA is a tracer mass balance represented as

$$c = \sum_{i=1}^n (q_i \times f_i) \quad (1)$$

where c is the concentration of a tracer in the aggregate sample, i is an index for sources, n is the total number of sources, q is the fraction of nitrate originating from a source, and f is the tracer concentration in a source sample. To test the validity of applying EMMA to our isotopic data set, we used a frequentist approach to generate numerous mixing polygons, which indicate the proportion of proposed end-member combinations that have a valid solution (Smith et al., 2013). The EMMA approach was deemed valid if all observed data points fell within the outermost mixing polygon.

Equation 1 can be applied to multiple tracers (e.g., $\delta^{15}\text{N}_{\text{NO}_3}$ and $\delta^{18}\text{O}_{\text{NO}_3}$) and is further constrained by an additional source mass balance

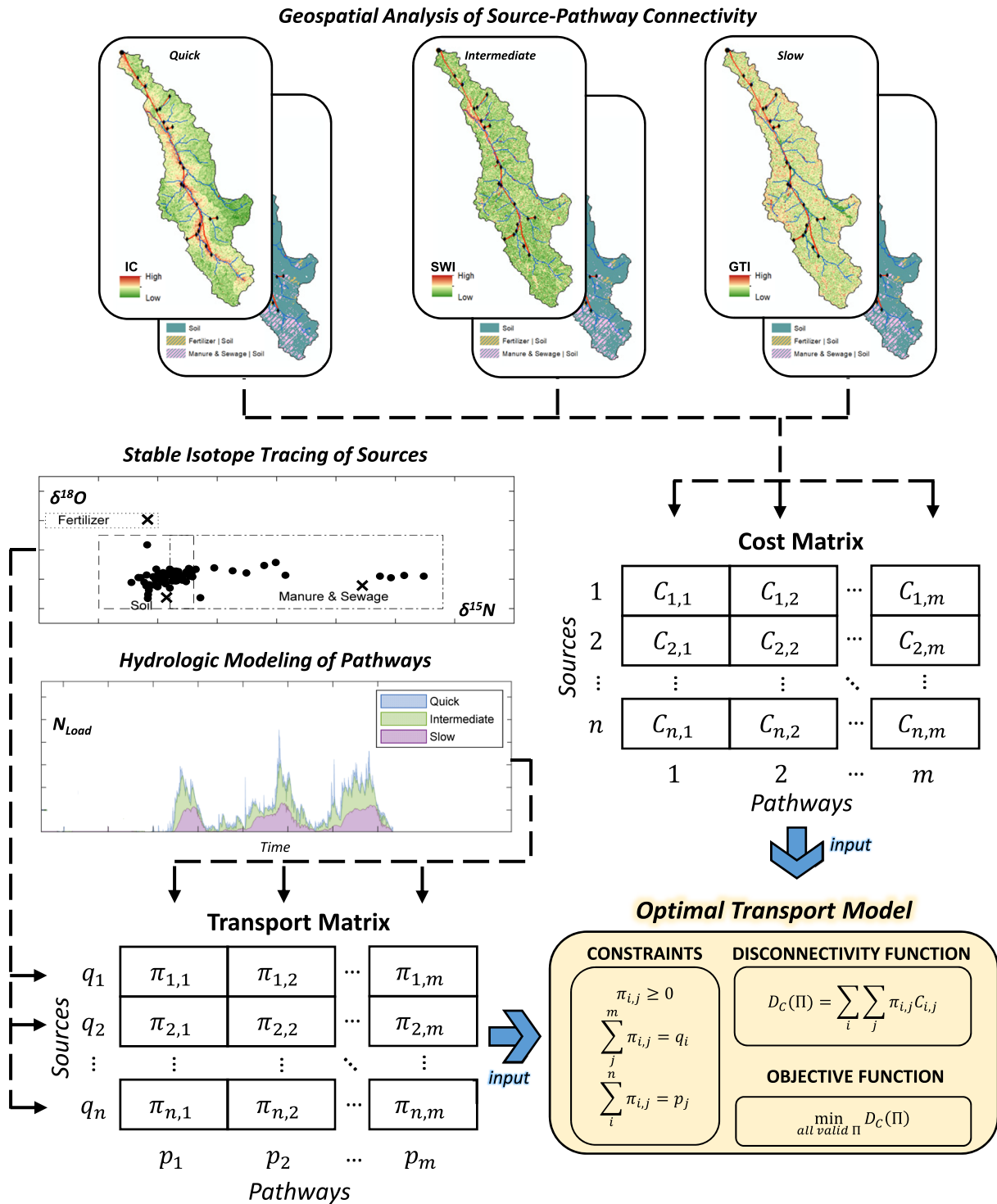


Figure 3. Schematic of optimal transport application to nutrient source-pathway connectivity. The transport matrix (π) describes the distribution of n sources to m pathways. The connectivity matrix (C) describes the spatial linkage of sources to their associated pathways. The optimal transport problem is constrained by an objective function which aims to minimize the global work performed and identify an optimal transport plan (Π) of source-pathway couplings.

$$\sum_{i=1}^n q_i = 1 \quad (2)$$

Likewise, there is also a mass balance constraint for multiple hydrologic pathways.

$$\sum_{j=1}^m p_j = 1 \quad (3)$$

where j is an index for pathways, m is the total number of pathways, and p is the fraction of nitrate transported along a pathway. In this study, the fractional contribution of pathways is known continuously through time from an existing calibrated hydrologic and nitrogen model of the Royal Spring basin (Husic, Fox, Adams, Ford, et al., 2019).

To describe the fractional contribution of nitrate sources apportioned to specific hydrologic pathways, we update the EMMA formulation to consider a joint-distribution mass balance (e.g., fertilizer nitrate transported by the intermediate pathway) as

$$\sum_{i=1}^n \sum_{j=1}^m \pi_{i,j} = 1 \quad (4)$$

where $\pi_{i,j}$ is the fractional contribution of source i across pathway j . The updated formulation is now unconstrained for anything more than just a single pathway. For example, in the case of three pathways ($m = 3$) and three sources ($n = 3$), six new unknowns are introduced, and the system of equations become unconstrained. This limitation has prevented researchers from ascribing source loading to specific pathways without the need for complex distributed modeling. We address this problem with optimal transport theory and resolve this limitation.

An optimal pairing of sources and pathways is one that minimizes the disconnectivity of a system (Figure 3). Minimizing disconnectivity is a logical objective function as it has parallels to work and kinetic energy minimization, which underscore many physical and ecological processes. A source-pathway pairing (π) represents how much of source mass q in the i th position is transported via pathway p in the j th position. A transport plan (Π) is a set of source-pathway pairings that satisfy the joint-distribution mass balance constraint (Equation 4). Disconnectivity (D_C) is calculated to be the total cost, or work required, of redistributing mass in the system

$$D_C(\Pi) = \sum_{i=1}^n \sum_{j=1}^m \pi_{i,j} C_{i,j} \quad (5)$$

where C is the cost of transporting mass from i source via j pathway. The above equation has two further mass balance constraints, where the sum of each row must equal to its source contribution and the sum of each column must equal to its pathway contribution

$$\sum_{j=1}^m \pi_{i,j} = q_i \quad (6)$$

$$\sum_{i=1}^n \pi_{i,j} = p_j \quad (7)$$

We parameterized the cost matrix using geomorphometric connectivity indices because of their ability to integrate factors that influence hydrologic transport (e.g., contributing drainage area, local topographic slope, and hydraulic conductivity) into a single value (Figure 3). In most optimal transport applications, the cost of transport is typically the Euclidean distance between masses, but here we modify the cost to also include topographic and geomorphologic characteristics to represent how arduous it is for a nutrient particle to traverse a distance. We assumed that the more well-connected particles (i.e., those located in regions with higher slopes and hydraulic conductivities) have a lower cost of transport relative to locations with low slopes or conductivities, such as depressions or aquitards.

For the quick flow pathway ($j = 1$), we used the IC to describe the connectivity of runoff to the basin outlet (Borselli et al., 2008; Cavalli et al., 2013) as

$$IC_k = \log_{10} \left(\frac{D_{up,k}}{D_{dn,k}} \right) = \log_{10} \left(\frac{\bar{W}_k \cdot \bar{S}_k \sqrt{A_k}}{\sum_{l=k}^{n_k} \frac{d_l}{W_l \cdot S_l}} \right) \quad (8)$$

where the subscript k indicates that each grid cell of the basin has its own IC value, n_k is the number of grid cells, $D_{up,k}$ is the weighted upslope component for routing of particles produced upgradient, $D_{dn,k}$ is the weighted downslope component for how far a particle must travel to reach a stream, $\bar{W}_k$ is the average weighing factor (herein the roughness factor; Cavalli et al., 2013), $\bar{S}_k$ is the average slope gradient of the upslope contributing area, A_k is the upslope contributing area, d_l is the length of the l th cell along the downslope path, W_l is the weight of the l th cell, and S_l is the slope of the l th cell. All geospatial properties were calculated using a 2 m LiDAR-derived digital elevation model (DEM) downloaded from the Geospatial Data Gateway (NRCS, 2018). The IC map was generated using the *SedInConnect* open source tool with the surface channel and subsurface conduit as the targets for downgradient connectivity (Crema & Cavalli, 2018). Larger IC values indicate that land parcels are more well connected to receiving downstream waterbodies (e.g., streams or springs).

For the intermediate flow pathway ($j = 2$), we used the Soil Topographic Wetness Index (SWI), an extension of the Topographic Index (Ambroise et al., 1996), to describe the wetness (connectivity) of shallow subsurface pathways (Anderson et al., 2014; Güntner et al., 2004) as

$$SWI_k = \ln \left(\frac{A_k}{D_k \tan S_k} \right) = \ln \left(\frac{A_k}{C_k H_k \tan S_k} \right) \quad (9)$$

where, for each cell, D_k is the soil transmissivity, C_k is the saturated hydraulic conductivity of the soil, and H_k is the soil depth to a restrictive layer. Soil properties for the study site were downloaded from the SSURGO database (NRCS, 2018). This approach considers the drainage area, local slope, and transmissivity properties of a soil for connecting water and solutes downgradient. Lower SWI values represent efficiently drained soils whereas larger SWI values indicate that the land parcel is likely to pond and generate saturation excess overland flow (Walter et al., 2002).

For the slow flow pathway ($j = 3$), we extend the SWI formulation to consider groundwater aquifer saturation, herein termed the Groundwater Topographic Index (GTI). We reasoned that this extension is viable as the Topographic Index has been shown to be a good predictor of shallow groundwater level (Anderson et al., 2015) and we calculated GTI as

$$GTI_k = \ln \left(\frac{A_k}{T_k \tan S_k} \right) = \ln \left(\frac{A_k}{K_k B_k \tan S_k} \right) \quad (10)$$

where, for each cell, T_k is the aquifer transmissivity, K_k is the saturated aquifer hydraulic conductivity, and B_k is the thickness of the aquifer. The active aquifer circulation zone/thickness is 15 to 60 m in the Inner Bluegrass (Scanlon, 1990) and we assumed an average value of 20 m. Saturated hydraulic conductivity data were compiled from studies in the region for limestone and alluvium (Palanisamy & Workman, 2014; Scanlon, 1990) and from the literature for limestone with interbedded shale (Kleeschulte and Seeger, 2003). This approach considers the drainage area, local topographic gradient, and transmissivity properties of the aquifer for connecting water and solutes downgradient. Lower GTI values represent highly transmissive bedrock whereas larger GTI values indicate that a land parcel is situated within an aquitard.

Transport cost (C) for a source-pathway coupling was calculated by normalizing all indices ($j = 1 \dots m$) between 0 and 1 and then calculating the spatial median for each source's contributing area ($i = 1 \dots n$). Thereafter, the indices were formulated so that greater connectivity equates to lesser cost as

$$C_{i,j} = \left[1 - \widetilde{IC}_i, \widetilde{SWI}_i, \widetilde{GTI}_i \right] \quad (11)$$

where $\widetilde{IC}_i$, $\widetilde{SWI}_i$, and $\widetilde{GTI}_i$ are the spatial medians (i.e., the most commonly occurring values) of the IC , SWI , and GTI , respectively, for each i th source. Selection of nitrogen source contributing areas was based

on National Land Cover Database (NLCD) land use (MRLC, 2019) and decades of experience studying contamination in the Royal Spring basin (Husic et al., 2020, 2017; Thrailkill et al., 1991; UKCAFE, 2011). We assumed the fertilizer source was most likely associated with cultivated crops, manure and sewage source to septic leakages and overflows associated with urban development, and soil nitrogen was considered for all land covers, including cropland, urban, forest, grassland, and open spaces.

The source-pathway connectivity problem has a near-infinite set of solutions (Figure 3 and Equations 5, 6, and 7); however, there is only a single transport plan (Π) that minimizes the total disconnectivity of the system. We assumed that the natural system will take the most energetically favorable redistribution of mass to reduce the total work performed (i.e., disconnectivity). This “optimal transport plan” provides a unique solution to our otherwise underdetermined system. This optimization has the following form

$$\min_{\text{all valid } \Pi} D_C(\Pi) \quad (12)$$

This is the Kantorovich formulation of transporting discrete masses, represented as a linear program, and we used a simplex algorithm (adapted from publicly available code; Peyré, 2018) to solve the problem iteratively until convergence (Peyré & Cuturi, 2019). All algorithms used in this study were coded into MATLAB 2018b and run on a Dell XPS 8930 with 32 GB of RAM.

2.4. Testing Impacts of Heterogeneity and Land Management

The cost matrix, derived from connectivity indices, has the effect of routing the nitrate load toward preferential couplings (i.e., heterogeneous connectivity). We test the extent of this preferential routing by comparing results to the assumption that all sources are equally connected to all pathways (i.e., homogeneous connectivity). The assumption of homogeneity assumes no a priori information about source-pathway coupling. To model homogeneous connectivity, each pathway's unmixing fraction was proportionally distributed to each source's unmixing fraction. For example, if the pathway model indicates 25% quickflow and our source isotope unmixing model shows 40% fertilizer, the fertilizer-quick source-pathway contribution would be 10% ($=0.25 \times 0.40$). In this method, there is no preferential (i.e., cost-reducing) distribution of mass along any source-pathway combinations, but rather all masses are proportionally distributed.

We also used our optimal transport model to virtually test the net effect of two land management scenarios: (1) implementing field buffers on agricultural cropland and (2) repairing failing septic systems in urban lands. In the first scenario, field buffers installation simulates the impact of reducing the fertilizer-quick source-pathway coupling. In the second scenario, reduced septic failure has the net impact of mitigating the importance of the manure-intermediate source-pathway coupling. To reflect the effects of land management, we adjust the transport cost matrix based on the intended result, which would be analogous to making a source less connected (i.e., managing it). For the first scenario, we set the fertilizer-quick cost ($C_{1,1}$) to a maximal value of “1.” Likewise, for the second scenario, we set the manure-intermediate cost ($C_{3,2}$) to “1” and assess the impact to the source-pathway connectivity matrix. To calculate the simulated reduction in nitrate loading, we take the difference in the proportional contribution of the targeted source-pathway coupling (e.g., fertilizer-quick in Scenario 1) before and after implementing the land management (i.e., raising the cost matrix value). Thus, the source-pathway matrix for the land management scenarios does not sum to 1 as the difference between its sum and unity is the amount of nitrate removed by land management.

2.5. Probabilistic Uncertainty Estimation

It is important to note that uncertainties in the respective source and pathway parent models will propagate into the optimal transport framework. As such, watershed modelers should take care to accurately identify and characterize sources and pathways prior to implementing the source-pathway model. First, regarding source uncertainty, we apply a Bayesian method, where posterior inference is determined by an importance sampling algorithm. The Bayesian model is adapted from ecological stable isotope mixing work and a full description of the framework can be found in Erhardt and Bedrick (2013). Isotopic signatures of the three end-member sources were estimated from published data collected from watersheds with similar nitrate sources to our own (Table 1). Isotopic enrichment factors for the three sources in the unmixing model were derived from the literature (Table 1). With regard to pathway uncertainty, the

Table 1
Nitrate Source End-Member Statistics Used for Bayesian Source Unmixing

Source	$\delta^{15}\text{N}$ (‰)	$\delta^{18}\text{O}$ (‰)	ε (‰)	Reference
Fertilizer ($N = 6$)	$+4.1 \pm 0.5$	$+20.4 \pm 3.0$	$+3.9 \pm 2.3$	Li et al. (2019); Zhang, Yang, et al. (2018)
Soil ($N = 6$)	$+5.7 \pm 2.0$	-6.2 ± 0.4	-4.1 ± 1.4	Liu et al. (2006); Zhang, Yang, et al. (2018)
Manure ($N = 5$)	$+22.2 \pm 2.2$	-2.1 ± 0.6	-0.6 ± 1.2	Li et al. (2019); Zhang, Yang, et al. (2018)

Note. “ N ” is equal to the number of end-member samples used in generating statistics. ε represents isotopic enrichment factors for each source. Values shown are the mean $\pm$ one standard deviation

nonprobabilistic Generalized Likelihood Uncertainty Estimation (GLUE) method was used on the lumped hydrologic and nitrogen model (Husic, Fox, Adams, Ford, et al., 2019), which provides concentration and load inputs to the connectivity framework. From that modeling analysis, an ensemble average of all 1,382 acceptable model realizations was used in this study to assess the variability in quick, intermediate, and slow pathway contributions. We outline the structure of the model and the analysis of uncertainty with the following pseudocode:

Begin

load pathway model data set ($N = 1,382$)

load stable isotope source data set ($N = 70$)

load $\widetilde{IC}$, $\widetilde{SWI}$, and $\widetilde{GTI}$ from geospatial connectivity maps

Algorithm 1 Optimal transport model

for $a = 1, \dots, X$ (number of runs to generate statistics)

randomly sample source data set

sample ensemble mean of pathway data set

Algorithm 2 Bayesian unmixing model

calculate source (q) fractions

calculate pathway (p) fractions

Algorithm 3 Linear transport simplex

repeat

generate Π

calculate $D_C(\Pi)$

until $\min D_C(\Pi)$

save Π

end for

calculate statistics

Finish

3. Results and Discussion

3.1. Nitrate Pathways, Sources, and Connectivity

Nitrate samples collected during the 30 month projected simulation period (2016 to 2019) showed good agreement with historical model calibration (2012 to 2016) (Figure 4a). The 2012–2016 results are identical to Husic, Fox, Adams, Ford, et al. (2019) while the simulations for 2016–2019 are presented for the first time. Sixty of the 70 new samples (85%) fell within modeled bounds. The good coverage of the data is particularly encouraging given that five highly sampled (subdaily to hourly) storm events, including an atmospheric river and a tropical cyclone (Husic, Fox, Adams, Backus, et al., 2019), were included in the data set. Hydrologic and biogeochemical conditions can change significantly at the spring during these extreme events (Husic, Fox, Adams, Ford, et al., 2019), which gives confidence to the robustness of the model. Further, the data also have well-balanced seasonal coverage with 17, 15, 16, and 22 samples collected during spring, summer, fall, and winter seasons, respectively.

3.1.1. Pathways

The pathways of nitrate transport in Royal Spring include quick, intermediate, and slow flow which represent conduit, fracture, and matrix flow paths, respectively (Figure 4b). Pathway modeling results showed

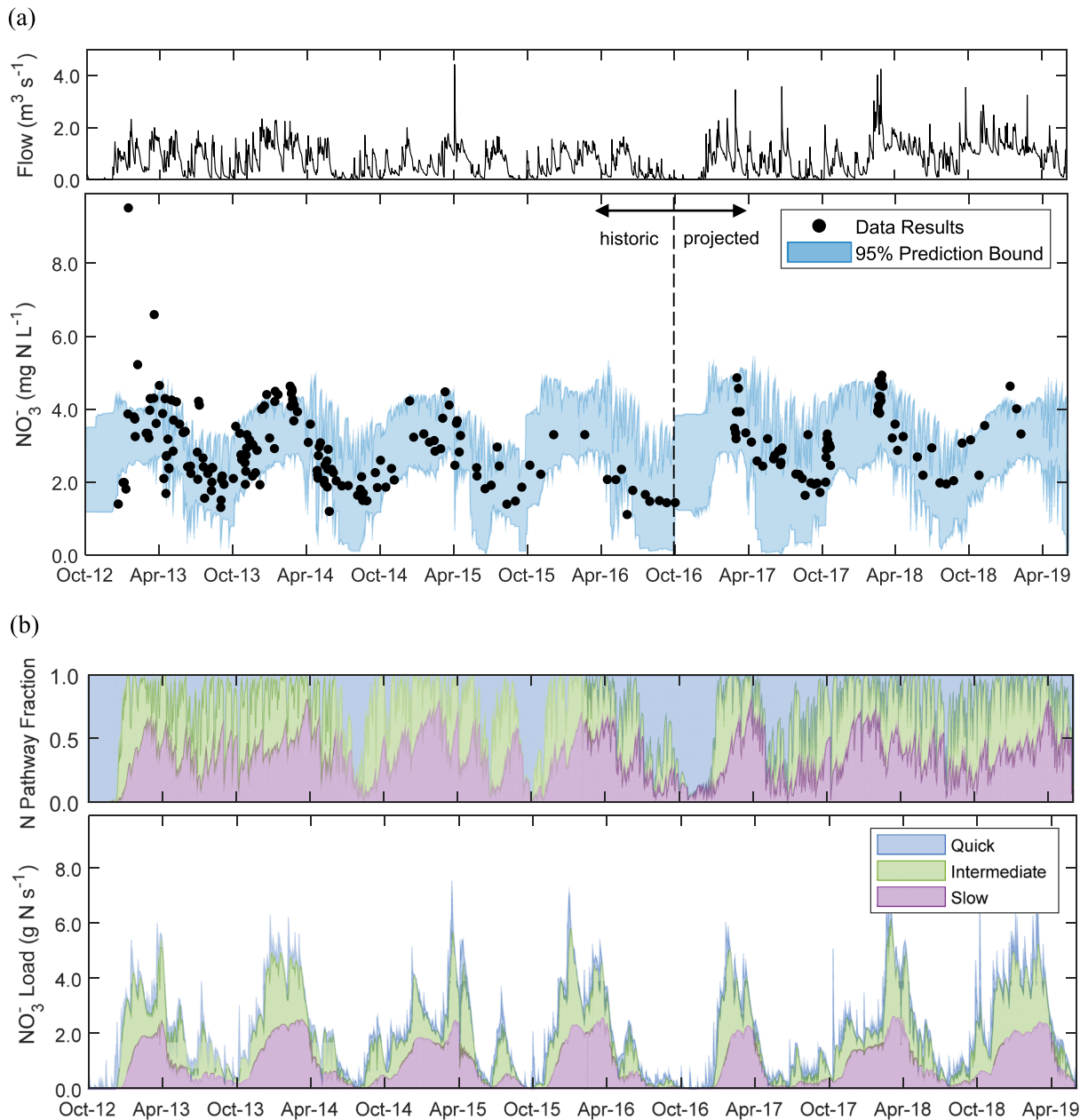


Figure 4. (a) Recorded discharge and modeled historical plus projected nitrate at Royal Spring from Husic, Fox, Adams, Ford, et al. (2019). The historical model runs from October 2012 to October 2016 ($n = 164$), and the projected model runs from October 2016 to June 2019 ($n = 70$). Both the historical and projected model simulations include most of the observed data, highlighting model robustness. (b) Ensemble average pathway modeling results from Husic, Fox, Adams, Ford, et al. (2019).

three distinct patterns in Royal Spring nitrate loading: (1) consistent slow flow contribution throughout the year, (2) increased importance of the epikarst and soil during medium-to-high flows as expressed by a greater intermediate flow contribution, and (3) large quick flow contribution during dry periods. To the first point, phreatic baseflow supplies water and nitrate to the spring during most of the year except for the occasional dry period with no spring flow. The relative importance of slow flow increases from October to April of each year, reflecting wetter conditions in the winter and spring. Thereafter, baseflow decreases as precipitation totals fall during the summer and pumping from the aquifer increases to sustain agriculture. Secondly, the intermediate/epikarst flow pathway contributes a high fraction of nitrate during the initial rising limb of the annual loadograph and during medium-to-high events. This

rising limb typically happens during the fall (between October and December) when the soil and epikarst zones are reconnected to the subsurface conveyance channel and drain material which has accumulated during the dry summer season. This “flushing” pattern is often observed in nonkarst systems as well where a rise in the water table reconnects stored material to transport pathways (Benettin et al., 2020; Botter et al., 2020). Lastly, quick flow contribution only dominates during dry, baseflow periods when the magnitude of nitrate loading is relatively low to the rest of the year. During limited baseflow, rapid influx of concentrated recharge can quickly dominate the spring load and even reverse the conduit-bedrock potentiometric surface until additional diffuse recharge reaches the groundwater table and reverts the system to the original potentiometric configuration (Vesper & White, 2004). This extensive modeling of karst pathway hydrology is crucial for understanding how unique sources are connected, mobilized, and transported along multiple pathways.

The temporal domain of when pathways are likely connected is highlighted by our load separation analysis (Figure 4b). Mean modeling results indicate that quick pathways are connected to the spring 38.5% of the year, intermediate 70.8%, and slow flow 70.0% of the year. Connection is established when a pathway load exceeds the observational accuracy of the model (i.e., the 10th percentile of N loading results). The observational accuracy of the model is imposed by daily pumping interference directly uphill of the spring gage (Husic, Fox, Adams, Ford, et al., 2019). This temporal pathway contribution is plausible as residence time analysis of our modeling framework shows short residence in quick pathways (3 ± 2 days), but longer residence times and, implicitly, transfer in intermediate (47 ± 10 days) and slow (122 ± 9 days) pathways (Husic, Fox, Adams, Ford, et al., 2019). The extent to which transport occurs along these hydrologic pathways is heavily influenced by mean transit times (Benettin et al., 2020; Lutz et al., 2018). For example, since sinkholes and swallets pirate surface water fairly quickly, the time duration of quickflow connection is similar to that of surface runoff, which has been approximated to be no more than 2 days in the region (Mahoney et al., 2018). On the other hand, subsurface fracture and matrix networks can become saturated and their flow paths connected for much longer durations (Ford et al., 2019). This result indicates that functional (dis)connectivity of nitrate pathways is temporally variable since connectivity of nitrate source pathways is mediated hydrologically (Ali et al., 2018; Zingaro et al., 2019).

3.1.2. Sources

Stable isotope data collected over the 30 month study period ($n = 70$) had mean $\delta^{15}\text{N}_{\text{NO}_3}$ and $\delta^{18}\text{O}_{\text{NO}_3}$ values of $+7.46 (\pm 5.09)\text{‰}$ and $+0.83 (\pm 2.67)\text{‰}$, respectively (Figure 5a). Most of the samples are near the soil nitrogen end-member and fall along the denitrification trend line although a few are in the manure and sewage range. All highly enriched manure and sewage samples ($n = 4$) were collected during winter months (between December and February). A linear regression of the remaining 66 samples indicates that some denitrification is present with a $\delta^{15}\text{N}_{\text{NO}_3}$ – $\delta^{18}\text{O}_{\text{NO}_3}$ slope of 0.37 ($p < 0.001$). Systems experiencing considerable denitrification typically have a slope between 0.43 and 1.0 (Granger & Wankel, 2016; Hu et al., 2019). Further evidence for limited denitrification is provided by the lack of a significant negative relationship between $\ln[\text{NO}_3^-]$ and $\delta^{15}\text{N}_{\text{NO}_3}$ or $\delta^{18}\text{O}_{\text{NO}_3}$ (not plotted), which is expected when NO_3^- is reduced as $\delta^{15}\text{N}_{\text{NO}_3}$ is enriched (Kendall et al., 2007). Thus, our results indicate that while biogeochemical processing (e.g., denitrification) may play a role, there is likely a large imprint of physical source-mixing that influences NO_3^- composition (Lutz et al., 2019).

Isotopic data results show that Royal Spring primarily drains a mixture of three nitrate sources: fertilizer, soil, and manure and sewage (Figure 5a). These sources are logical as Royal Spring is a mixed land use watershed that drains some of the greatest density of horse farms in the world, a large fraction of agricultural land, and a populous downtown area plagued by frequent combined storm sewer overflows (UKCAFE, 2011). We assume no major contributions from other end-members, such as NO_3^- and NH_4^+ in precipitation, as none of the 70 samples we collected fell in the expected range of those sources. A mixing polygon analysis, generated from 10,000 Monte Carlo simulations, provides quantitative support for applying an isotopic mass balance approach to this data set (Figure 5b). All data points are bound by the outermost mixing polygon, indicating that each data point is a plausible combination of the selected end-members (Smith et al., 2013).

Quantitative Bayesian unmixing indicated that soil nitrogen ($51.2 \pm 17.4\%$) is the dominant source of spring nitrate followed by fertilizer ($27.8 \pm 9.2\%$) and then manure ($21.0 \pm 19.6\%$). The large soil nitrogen

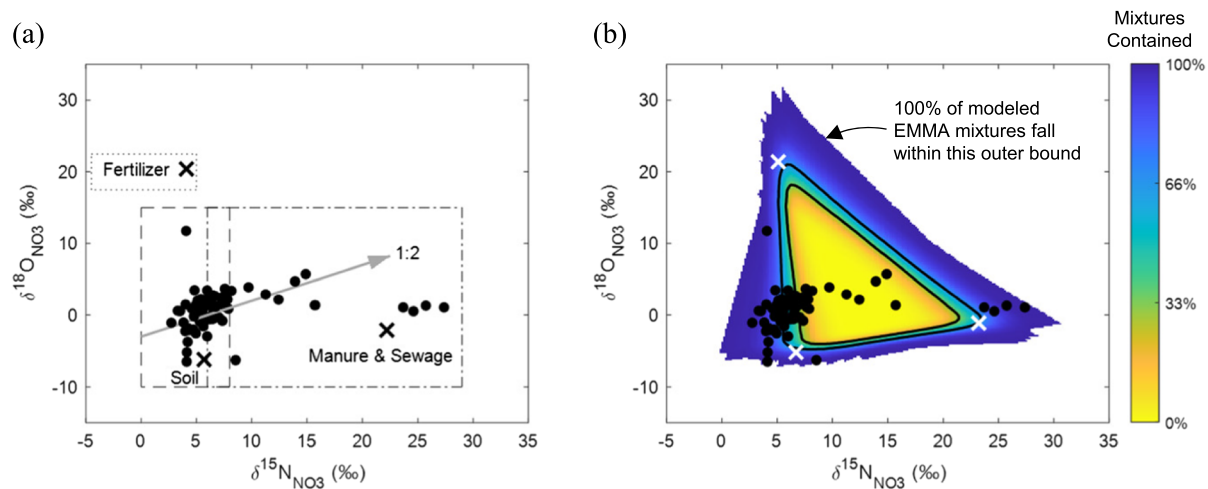


Figure 5. (a) Nitrate isotope biplot. Most samples (circles, $n = 70$) fall near the soil end-member and along the denitrification trend line, although several are in the manure and sewage range. The theoretical denitrification trend line (with a 1:2 slope) represents the expected progression of $\delta^{15}\text{N}_{\text{NO}_3}$ and $\delta^{18}\text{O}_{\text{NO}_3}$ as denitrification progresses. Unmixing end-members (x-marks, $n = 3$) are estimated from the literature (Table 1). (b) Mixing polygon representing the percentage of EMMA mixtures contained within the possible solution space. All possible solutions are contained within the outermost bound with the solution space shrinking further into the polygon.

contribution is likely derived from mineralization of organic compounds and potentially nitrification of ammonium (Kendall et al., 2007). While synthetic fertilizer, on average, contributes more to spring nitrate than manure and sewage, fertilizer only makes up a majority (>50%) of the spring nitrate signal less than 1% of the time. On the other hand, manure and sewage comprises over half of spring nitrate in 10% of all samples, highlighting that there is substantial heterogeneity in manure and sewage connectivity. Sources that are activated less often but contribute disproportionately to downstream loading may be more suitable for land management if their spatial extent is limited and the source can be disconnected from transport (Jarvie et al., 2017). In the context of our results, legacy nitrogen within the soil is a dominant contributor to spring nitrate, and the principle method for reducing this loading would be through planting of cover crops. However, recent research has indicated that even with 100% efficiency of nitrogen use in soils, it would take decades for rivers in the United States to meet quality standards (Van Meter et al., 2018), indicating that new perspectives on nitrogen management are needed to achieve water quality goals. With confidence in the pathways and sources of nitrate as well as the applicability of unmixing methods, we next explored the strength of connectivity between hydrologic pathways and nitrogen sources.

3.1.3. Connectivity

Geospatial analysis of Royal Spring highlights spatial heterogeneities in the location of strongly connected hydrologic pathways and the availability of dominant nitrogen sources (Figure 6). For quick pathways, higher IC values were associated with sloping hills that flank the primary surface channel. Likewise, lower IC values correspond with parts of the landscape distal to the mainstem pathway. Most of the mapped swallets (i.e., sinkholes within streams) are located along the main corridor of the stream network and connect to the subsurface conduit (Husic et al., 2017), giving us confidence that IC accurately represents quick flow connectivity. For intermediate pathways, lower SWI values, which we interpret as more well-connected soil zones, were in areas where the topographic gradient and soil hydraulic conductivity were largest. Higher SWI values were associated with lowland areas susceptible to the accumulation of soil moisture and generation of overland flow, such as those with shallower soil depths and lower hydraulic conductivities. For slow pathways, GTI values have greater variability compared to SWI values because aquifer hydraulic conductivity is more variable than soil hydraulic conductivity in the Royal Spring basin. Spatial median IC values indicate that agricultural landscapes (i.e., fertilizer) are most well-connected to quick pathways (Table 2), whereas SWI and GTI median values indicate that manure and sewage are most well-connected to intermediate and slow pathways, respectively. Note, however, that the absolute value of an index is not what is important to forthcoming model predictions. Rather, what is important is the relative difference between any two index values. For example, model simulations using

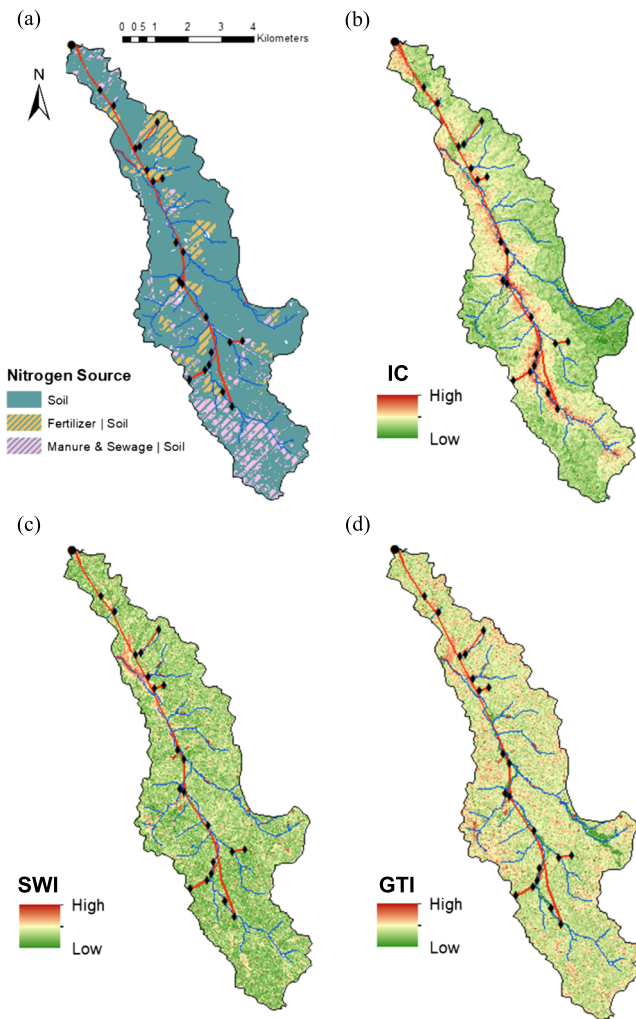


Figure 6. Geospatial analysis of Royal Spring (a) dominant nitrogen sources, (b) Index of Connectivity (*IC*); (c) Soil Topographic Wetness Index (*SWI*); and (d) Groundwater Topographic Index (*GTI*).

Table 2
Transport Cost (C_{ij}) Matrix

	IC	SWI	GTI
Fertilizer	0.797	0.275	0.367
Soil	0.801	0.266	0.355
Manure	0.818	0.250	0.340

Note. “1” represents maximal cost, and “0” represents minimal cost. Each cost was calculated as the median value of a pathway’s index that overlies a source’s contributing area. The indices are *IC* = Index of Connectivity (quick), *SWI* = Soil Topographic Wetness Index (intermediate), and *GTI* = Groundwater Topographic Index (slow)

IC-fertilizer, *IC*-soil, and *IC*-manure equal to 0.797, 0.801, and 0.818, respectively, provide the exact same results as when all three indices are offset equivalently (e.g., 0.497, 0.501, and 0.518).

We have confidence in the selection of these geomorphometric indices to represent landscape connectivity and cost of transport due to their integration of dominant hydrologic processes, such as contributing drainage area, local topographic slope, and hydraulic conductivity (Buchanan et al., 2014; Cavalli et al., 2013). The *IC* metric is advantageous as it can easily be generated from the *SedInConnect* tool (Crema & Cavalli, 2018), only requires a DEM data set, and has been met with success in a variety of settings, including in lowland and mountainous systems (Gay et al., 2016; Mishra et al., 2019). Alternative connectivity metrics for surface flow include the Topographic Wetness Index and its extension the Network Index, which considers wetness along a continuous path from source to sink (Lane et al., 2009). However, we utilize *IC* rather than these indices because of its ability to consider microtopography and its general ubiquity in the connectivity literature (Cavalli et al., 2013). To the authors’ knowledge, there are presently few quantitative index alternatives to the *SWI* for understanding subsurface connectivity (Blume & van Meerveld, 2015). We have confidence in the *SWI* and *GTI* as their theoretical constructions include drivers of subsurface flow, such as the hydraulic conductivity, drainage area, and hydraulic gradient (from kinematic approximation of topographic slope). Further, field verifications of the *SWI* have shown that it is a stronger indicator of subsurface saturation than the Topographic Wetness Index in saturation excess settings (Güntner et al., 2004; Walter et al., 2002). With an understanding of how well-connected particles in the landscape are to the watershed outlet, we utilize this information in the next section as a quantitative proxy for the cost of transport.

3.2. Optimal Transport Model Application

3.2.1. Source-Pathway Connectivity Matrix

The results of our optimal transport model are presented as a source-pathway connectivity matrix (Figure 7). First, we investigate the case of optimal transport (heterogeneous connectivity), which considers that certain sources are preferentially connected to certain pathways (Figure 7a). Over the 30 month study period, the dominant source-pathway coupling was soil-slow ($27.2 \pm 14.9\%$) followed by fertilizer-quick ($16.4 \pm 11.5\%$) and manure-intermediate ($15.4 \pm 11.2\%$). The remainder of the nitrate load was transported by pathways of lesser import, from soil-intermediate ($14.0 \pm 14.6\%$) down to fertilizer-slow ($0.3 \pm 2.2\%$). Thus, the top three source-pathway couplings account for more nitrate export (59.0%) than the bottom six couplings combined (41.0%). The slow pathway transports predominantly soil nitrogen ($27.2 \pm 14.9\%$) with relatively little fertilizer ($0.3 \pm 2.2\%$) and manure ($7.9 \pm 13.3\%$). Likewise, the quick pathway primarily exports fertilizer ($16.4 \pm 11.5\%$) with very little soil nitrogen ($6.9 \pm 16.1\%$) and manure ($0.8 \pm 2.8\%$). On the other hand, the intermediate pathway dynamics were not dominated by a single source but rather included considerable loading by all three: fertilizer ($11.1 \pm 11.7\%$), manure ($15.4 \pm 11.2\%$), and soil ($14.0 \pm 14.6\%$). With an estimate of the magnitude of each source-pathway connection, we next utilize our optimal transport model results to infer internal hydrobiogeochemical processes in karst.

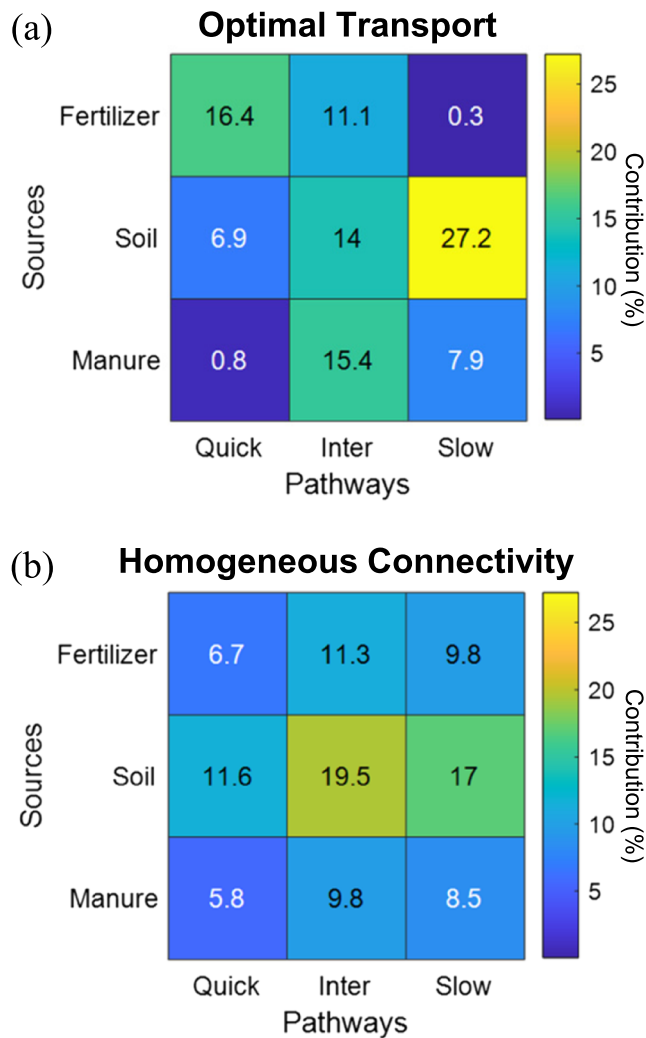


Figure 7. Source-pathway connectivity matrix for (a) optimal transport (i.e., certain sources preferentially connected to certain pathways) and (b) homogeneous connectivity (i.e., all sources equally connected to all pathways). The number inset within each tile represents the percentage of the total nitrate load at the spring ascribed to a source-pathway coupling.

transport, soil-slow, now comprises a substantially smaller fraction of the overall load ($17.0 \pm 19.8\%$). Further, source-pathway couplings that were considered negligible in the case of optimal transport now constitute an order of magnitude greater fraction of the nitrate load, such as fertilizer-slow (up from 0.3% to 9.8%) and quick-manure (up from 0.8% to 5.8%). With homogeneous connectivity, the bottom three source-pathway couplings transport 21.0% of the total nitrate load compared to just 8.0% in the case of optimal transport. Therefore, if homogeneity in source-pathway coupling were assumed, the limited resources available for nutrient management could potentially be misallocated on marginal couplings. For example, EMMA results indicate that $27.8 \pm 9.2\%$ of the nitrate load is derived from fertilizer, which can be addressed through a variety of land management strategies, such as timing of application, implementation of buffer strips, and planting of cover crops. If homogeneous connectivity were assumed, the source-pathway matrix would indicate that cover crops (i.e., a method to reduce fertilizer-intermediate loading) are the best management practice rather than field buffers (i.e., a method to reduce fertilizer-quick) as indicated by optimal transport. Thus, the inclusion of heterogeneity within the optimal transport model aides in improved decision-making by combing source, pathway, and connectivity modeling under one framework.

The optimal transport model and source pathway connectivity matrix highlights the dynamic nature of the karst critical zone (epikarst) to intercept, store, and transport considerable quantities of nitrate. Besides exporting the largest fraction (40.5%) of the total nitrate load of any pathway, intermediate flows were also composed of the most variable source composition. The epikarst is a near-surface water-adsorption zone subject to extreme weathering (Aquilina et al., 2006), and it underlays much of our study region. Thus, the epikarst is the first to intercept soil-water percolation, which can be highly concentrated in agricultural lands, as well as material that bypasses the soil through macropores and large fractures (Aquilina et al., 2006; Fretwell et al., 2005). Likewise, in our system, manure and sewage inputs from failing subsurface septic systems are directly injected into the epikarst and entirely circumvent surface pathways, such as runoff (Katz et al., 2010). Once material enters the epikarst, it can be stockpiled until hydrologic connectivity is restored and pathways are rewetted at which point considerable export occurs (Tritz et al., 2011; Z. Zhang, Chen, Cheng, & Soulsby, 2020) and can result in peak stream concentrations of nitrate (Husic, Fox, Adams, Backus, et al., 2019). Regarding the quick and slow pathways, representing runoff and base-flow, respectively, fertilizer likely dominates the quick pathway loading because crops in the region are often grown on gently rolling hills, which are suitable for row cropping, but also are susceptible to sinkhole formation in local depressions (Boyer & Pasquarell, 1995). Soil nitrogen constitutes a large fraction of the slow pathway likely due to leaching/percolation of soil nitrogen from the epikarst to the phreatic zone when the epikarst is disconnected from intermediate pathways (Husic, Fox, Adams, Ford, et al., 2019). Thus, any kind of land management could be further informed by the source-pathway connectivity matrix because it indicates not only what the dominant sources of contamination are, but also what pathways those sources take to downstream waterbodies.

3.2.2. Impacts of Source-Pathway Heterogeneity

While heterogeneous connectivity results in the concentration of the nitrate load toward preferential couplings, homogeneous connectivity assumes that all sources are equally connected to all pathways (Figure 7b). In this case, the magnitudes of dominant source-pathway couplings are mitigated while marginal couplings are given greater importance. The most dominant pathway in heterogeneous (optimal)

The homogeneous assumption is also problematic in the context of process representation, with studies highlighting that hydrological transport and cycling can disproportionately occur within small areas of karst watersheds (Hartmann, 2016; Hartmann et al., 2017). To this extent, not all pathways are created equal, and our results indicate stark variability in how sources are transported by hydrologically connected pathways. A primary reason that the heterogeneous (Figure 7a) and homogeneous (Figure 7b) source-pathway matrix solutions are so disparate is because of the karstic properties of Royal Spring. Pathway diameters and hydraulic conductivities of the geologic substrate can vary by orders of magnitude within small spatial extents, which makes modeling karst at the fine-resolution considerably challenging (White, 2002). By utilizing the physically grounded connectivity indices, averaged over different source contribution areas, our optimal transport model integrates this heterogeneity to a scale that makes it manageable. While the heterogeneity in the collocation of transportable mass and highly connected pathways is recognized as a dominant control on downstream water quality (Botter et al., 2020; Mahoney et al., 2018; Musolff et al., 2017; Zhi et al., 2019), existing methods have been limited in their ability to quantify the explicit impacts of heterogeneity (Wohl et al., 2019). We were able to quantify the explicit impacts of heterogeneity because the optimal transport model output provides a unique solution that minimizes the total work performed at the watershed-scale. This solution is then integrated into the source-pathway connectivity matrix where inferences regarding important couplings can be made and management decisions can be informed. From this, we deduced that the assumption of homogeneity is inadequate as it underestimates dominant couplings and overestimates contribution from marginally important couplings. This finding highlights the need for explicit consideration of heterogeneity in water quality models. Optimal transport theory has shown success when applied to image processing (Peyré & Cuturi, 2019), traffic networking (Santambrogio, 2015), and bird migration (Nichols et al., 2017), and our results complement this growing body of literature and draw attention to the utility of optimal transport to assess heterogeneity in nitrate source-pathway connectivity.

3.2.3. Applicability of the Optimal Transport Model

In fractured karst, with its extensive heterogeneity, distributed modeling of hydrology lags significantly behind that of porous media (Hartmann et al., 2014) and current numerical modeling tools struggle to adequately resolve how sources and pathways are coupled in karst settings. Typically, to resolve these limitations, a distributed numerical model would be required to model the pathways that a particular source takes to get to the stream environment (Z. Zhang, Chen, Cheng, Li, et al., 2020). The added value of our optimal transport modeling framework is that it allows land managers and researchers to resolve these limitations through the integration of commonly collected in-stream data sets (discharge, nitrate concentration, isotopic composition) with parsimonious modeling tools (lumped-reservoir models, loadograph separation, connectivity maps). While the results we generate can also be derived through distributed modeling, which resolves complex physics and biogeochemistry at a fine-scale, these other models are difficult to calibrate, require extensive input data sets, and are difficult to transfer from one setting to another without considerable alteration (Fatichi et al., 2016). By bringing together feed-forward conceptual (pathway) modeling with data-driven inverse (source) modeling and connectivity mapping, optimal transport allows us to tackle an emerging area of research that's at the center of the convergence of these methodologies (Jensen et al., 2018; Lutz et al., 2017; Z. Zhang, Yang, et al., 2018).

Beyond the exact methodological tools we use to execute the optimal transport model, there are many alternatives for generating the necessary source, pathway, and connectivity inputs (Table 3). For example, while a lumped hydrologic and nitrogen model that provides continuous estimate of pathway loading is robust (Hartmann et al., 2016; Husic, Fox, Adams, Ford, et al., 2019; Sullivan et al., 2019), there are less model-intensive alternatives such as loadograph separation (Fenton et al., 2017; Mellander et al., 2012; Miller et al., 2017). Loadograph separation only requires nitrate concentration and discharge, which are commonly becoming ubiquitous, particularly with the widespread use of aquatic nitrate sensors that provide continuous estimates of nitrate concentration (Burns et al., 2019). Further, with regard to nitrate sourcing, stable isotopes are commonly employed (Lutz et al., 2019; H. Zhang, Kang, et al., 2020; Wang et al., 2020; Z. Zhang, Chen, Cheng, Li, et al., 2020; Z. Zhang, Chen, Cheng, & Soulsby, 2020) but sources have also been unmixed with elemental concentration data alone (Katz et al., 2011; Yue et al., 2017). For example, the relationship between nitrate and chloride concentration provides a simple way to estimate whether nitrate is derived from wastewater or agriculture origin (Chen et al., 2009). Lastly, the determination of transport

Table 3
Model Inputs for the Optimal Transport Framework (Pathways, Sources, and Connectivity) and Alternative Methodological Tools to Derive the Inputs

Model input	Methodological tool(s)	References
Pathway loading	Lumped-reservoir model of hydrology and biogeochemistry	This study; Hartmann et al. (2016); Husic, Fox, Adams, Ford, et al. (2019); Sullivan et al. (2019)
	Physically based semidistributed models (e.g., SWAT)	Ikenberry et al. (2017); Lam et al. (2010); Singh and Stenger (2018)
	Loadograph recession analysis of high-frequency sensor solute data	Fenton et al. (2017); Mellander et al. (2012); Miller et al. (2017)
	Statistical age-based and sinusoidal filtering models	Benettin et al. (2020); Lutz et al. (2018); Zhang, Kang, Wang, Zhang, and Chen (2020)
Source unmixing	Stable isotope ($\delta^{15}\text{N}$ and $\delta^{18}\text{O}$) mixing models	This study; Lutz et al. (2019); Wang et al. (2020); H. Zhang, Kang, Wang, Zhang, and Chen (2020)
	Conservative solute (e.g., chloride) and nitrate mixing models	Chen et al. (2009); Katz et al. (2011); Yue et al. (2017)
	Classification trees and catchment budgets	Jawitz et al. (2020); Mattern et al. (2009); Spruill et al. (2000)
Transport cost	Geomorphometric connectivity indices	This study; Buchanan et al. (2014); Crema and Cavalli (2018); Lane et al. (2009)
	Stochastic likelihood of connectivity	Mahoney et al. (2018, 2020a, 2020b)

Note. Some referenced studies may utilize multiple listed methods.

cost can be done with geospatial mapping of connectivity indices (Buchanan et al., 2014; Crema & Cavalli, 2018; Lane et al., 2009) or with more rigorous stochastic methods that employ semidistributed models to account for spatiotemporal variability in connectivity (Mahoney et al., 2018, 2020a, 2020b).

3.2.4. Drivers of Spring Nitrate Loading

The source-pathway connectivity matrix is not stationary across discharge intensity (Figure 8a). In terms of pathways, slow flow is a prominent fraction of the total nitrate load during the lowest discharge quartile as it provides baseflow to the spring. Quick flow is also prominent because of flashy behavior at the spring during dry periods, which is a result we expected as the spring discharge typically responds the same day as rainfall inputs (Husic, Fox, Adams, Ford, et al., 2019). Likewise, during low flows when the epikarst is disconnected or unwetted, intermediate pathways only contribute a small fraction of the nitrate load. When the epikarst becomes wetted in the second quartile (i.e., hysteresis) (Tritz et al., 2011), it begins to be a dominant flow and nutrient pathway. Further, the epikarst is recognized to drain highly concentrated nitrogen from the soil-zone during high flows (Husic, Fox, Adams, Backus, et al., 2019), which further contributes to elevated spring loading. Slow flow gains prominence as baseflow contributions become greater (50–75% quartile), however it decreases at the highest discharge quartile as extreme events are heavily loaded by the quick and intermediate pathways (Husic, Fox, Adams, Backus, et al., 2019).

Regarding sources, soil nitrogen is a large contributor across all discharge quartiles, but its relative importance slightly decreases with increasing discharge ($p < 0.05$). Fertilizer contribution to the total spring nitrate load does not appear to be impacted by discharge conditions ($p = 0.71$), agreeing with our optimal transport matrix that showed fertilizer was mostly associated with the slow pathway (Figure 7a), which steadily

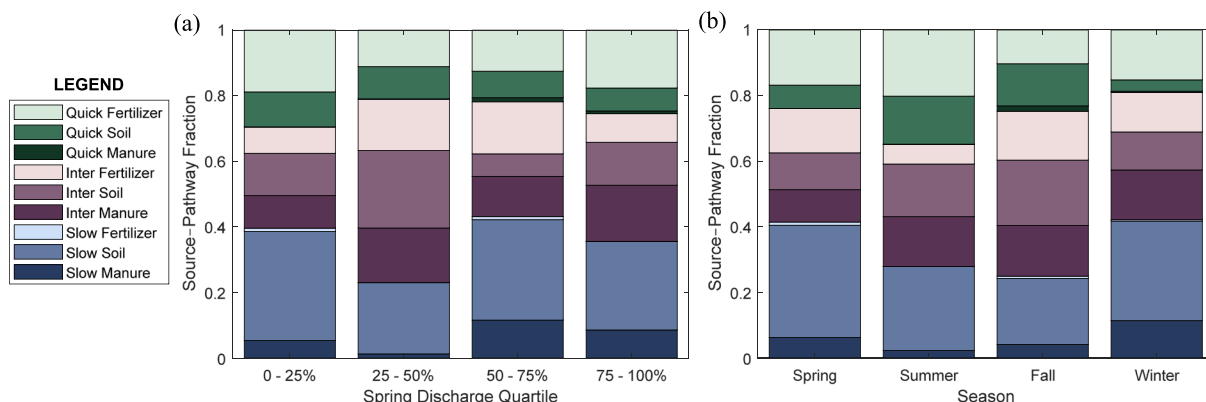


Figure 8. Mean source-pathway contribution as a function of (a) spring discharge and (b) seasonality.

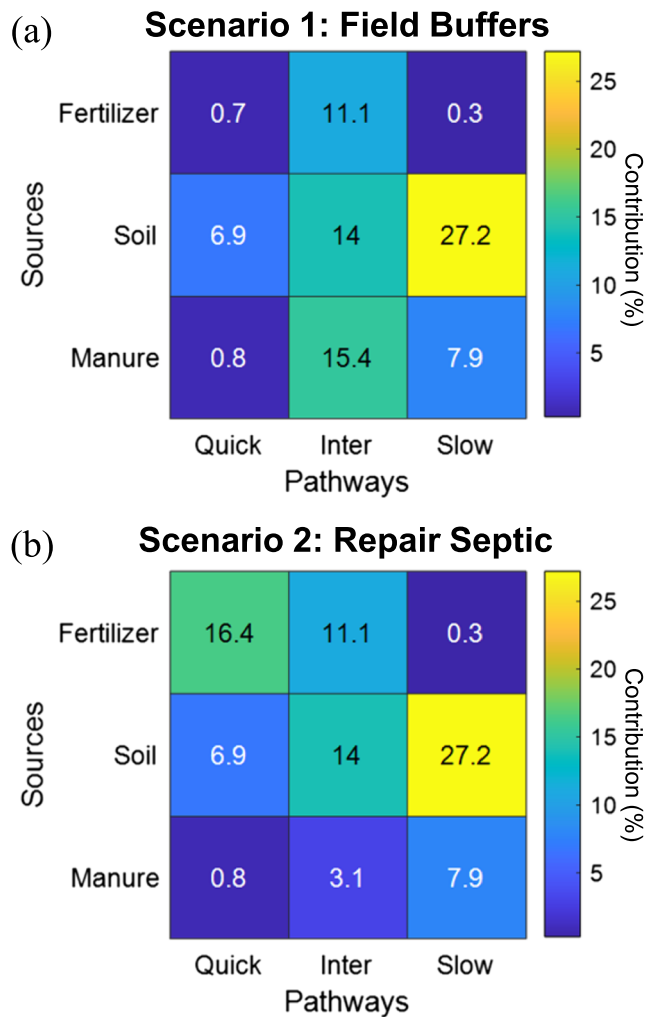


Figure 9. Source-pathway connectivity matrix for two scenarios that consider land management, including (a) implementation of field buffer strips in agricultural croplands, which mitigates runoff of fertilizer to quick pathways, and (b) repairing failing septic and sewer systems in urban lands, which mitigates shallow subsurface flow of manure and sewage to intermediate pathways. The number inset within each tile represents the percentage of the total nitrate load at the spring ascribed to a source-pathway coupling. Note: The matrices do not sum to unity as some nitrate mass has been removed as a result of land management.

contributes to the spring regardless of flow conditions (Figure 8a). On the other hand, the manure source becomes more prominent as discharge increases ($p < 0.05$) due to elevated intermediate pathway contribution during medium-to-high flows. This increase was also reflected in our stable isotope data that showed nearly all samples, highly enriched in $\delta^{15}\text{N}_{\text{NO}_3}$, indicating a manure source (Figure 5a), occurred during winter when flows were high relative to the rest of the year. However, regardless of the magnitude of discharge, our findings indicated that nitrogen derived from the soil zone is the primary source of nitrate delivered to Royal Spring. These findings are consistent with the high historical land use typical of the Bluegrass Region of Kentucky (Ford et al., 2019; Husic et al., 2020; Husic, Fox, Adams, Ford, et al., 2019; UKCAFE, 2011). The current study gives a new perspective on the coupling of nitrate sources with hydrologic pathways so that decision making is informed by physical and biogeochemical N processes as well as hydrologic conditions.

Lastly, the source-pathway connectivity matrix is also not stationary with respect to seasonality (Figure 8b). Again, quick flow is more prominent in the summer season when discharges are typically lower and summer thunderstorms cause the immediate influx of concentrated recharge (Vesper & White, 2004). Epikarst dominates the load during the fall when it is rewetted and can begin to drain nitrate that has built-up in the soil layer during the dry summer period (Husic, Fox, Adams, Ford, et al., 2019). This shows good agreement with surface water quality studies where large pulses of nutrients at the onset of the wet season are typically associated with agricultural land drainage (Midwest, USA; Lin et al., 2019) and flushing from shallow soils (Western France; Benettin et al., 2020). Further, nitrate not assimilated by plants during dry periods may accumulate in the soil and become more susceptible to leaching once wet weather occurs. Lastly, slow flow is most important during the wet season (winter and spring) because of higher precipitation, lower temperatures, and reduced pumping from the aquifer. The relative importance of baseflow N contributions during the wet season are widely reported from worldwide modeling and monitoring studies (see meta-analysis in Hu et al., 2019). Whereas other studies have linked nitrate and water contributions from groundwater to highlight the importance of slow flow pathways, we provide a method for connecting these loads to various sources.

3.2.5. Optimal Transport as a Management Tool

The optimal transport modeling framework allowed for the simulation of how different land management scenarios could impact the source-pathway connectivity matrix (Figures 9a and 9b). Based off the results of our optimal transport analysis (Figure 7a), we selected

fertilizer-quick ($16.4 \pm 11.5\%$) and manure-intermediate ($15.4 \pm 11.2\%$) for testing virtual land management scenarios. The predominant source-pathway coupling, soil-slow ($27.2 \pm 14.9\%$), was not selected for land management as it is uncertain how much of the coupling is background nitrate derived from soil origin versus anthropogenically augmented nitrate. To mitigate the magnitude of the two source-pathway couplings through human intervention we tested (1) implementing field buffers on agricultural cropland and (2) repairing failing septic systems in urban lands. In the first case, buffer placement would be located along streambanks and near mapped/suspected sinkholes to intercept fertilizer runoff and remove it from downstream transport. In the second case, a survey of septic system and storm sewer health would be conducted and tested for leaks. Sites with identified leaks would be repaired and the leaks sealed, thus cutting off the introduction of manure into the critical zone (epikarst). As a reminder, we test each scenario by adjusting the cost matrix to reflect the impedance of a targeted source-pathway coupling and quantify the reduced contribution of the coupling.

Under Scenario 1 (Figure 9a), which assumes that field buffer strips are planted in susceptible cropland zones, the contribution of fertilizer-quick is reduced from $(16.4 \pm 11.5\%)$ to $(0.7 \pm 2.9\%)$. Thus, the overall nitrate load is reduced by 15.7% in Scenario 1. This has implications for other pathways as fertilizer maintained on the crop-field has an increased chance of being taken up by crops and removed from transport (Van Meter et al., 2018), which is not accounted for by our model, but would yield a benefit to downstream aquatic health. Under Scenario 2 (Figure 9b), which assumes that failing sewer and septic systems are rehabilitated, the contribution of manure-intermediate is reduced from $(15.4 \pm 11.2\%)$ to $(3.1 \pm 10.2\%)$. Thus, the overall nitrate load is reduced by 12.3%. By preventing sewage inputs into the epikarst, the nitrate load can be successfully routed to wastewater treatment plants where it can be successfully remediated. Failed septic and sewer systems have been highlighted as one of the primary causes of degrading water quality at Royal Spring. So much so that the combined sewer overflow system in the urban headwaters of the Royal Spring basin are undergoing a \$600 million renovation to prevent leaching of waste into the soil and epikarst (UKAFE, 2011). These scenarios were performed to demonstrate how land management practices can be reflected as a further parameterization of the transport cost function beyond just the connectivity values derived from the geomorphometric index analysis.

The net effect of land management practices is an increase in the cost (i.e., reduction in connectivity) and, conversely, a decrease in the importance of a source-pathway coupling. Here we test hypothetical scenarios, but in study basins with more extensive geospatial data (Qiu, 2009), it may be appropriate to use that data to amend the transport cost function beyond just the connectivity index value to reflect land management alterations to source-pathway connectivity. Further, while we chose an arbitrary increase in the cost of transport (i.e., we set it to “1”), sites with the requisite data could inversely determine a transport cost based on observed load reductions. Likewise, other modeling tools, such as running Soil and Water Assessment Tool scenarios under different management strategies (Ricci et al., 2020), could be used as a comparison to the optimal transport results to validate the cost function and management efficacy. However, as legacy nitrogen can take decades to exit watershed systems (Van Meter et al., 2018), it is important to coincide management strategies that consider both the time scales of pathway transport and the connectivity and retention of source zones. Thus, our optimal transport model provides a framework for evaluating historic and projected land management scenarios and their net effect to altering the distribution of source-pathway loading.

4. Conclusions

We show that optimal transport theory can successfully be applied to nitrate stable isotope data, hydrologic modeling, and geospatial connectivity mapping to infer source-pathway connectivity. We also demonstrate that optimal transport theory allows for the solution of underdetermined problems in water quality modeling and provide a blueprint for other researchers and land managers to integrate their own data sets into the new model framework. Additionally, we highlight how spatial heterogeneity in the connectivity of sources and pathways shifts the dominant couplings of nitrate transport through space and time. Our analysis indicated that fertilizer carried by quick flow and manure carried by intermediate flow were two large contributors to the spring nitrate load. Lastly, we show how the optimal transport model can be utilized to simulate targeted land management strategies, such as planting field buffers and repairing failed septic systems, to mitigate nitrate contamination.

A limitation of the optimal transport model applied in this study is the spatiotemporal resolution of the inputs, particularly the nitrogen pathway model. An improvement for future studies would be to resolve the hydrology and biogeochemistry at finer scales, such as with semiphysically based models (e.g., SWAT) that combine the simplicity of lumped storage models with the rigor of physical hydrologic routing. Further, the present model operates at the daily time step so subdaily transient flushing of some material may not be captured by the temporal resolution of the model. The modeled daily time step here is imposed by a need to satisfy the Courant-Friedrichs-Lewy condition, but that time step could be reduced (e.g., from daily to hourly) as the spatial resolution of the model also becomes finer. Because of its relative parsimony, the presented optimal transport framework is easily transferable to other sites and can be integrated into large swaths of existing flow and nutrient data to generate more knowledge of source-pathway coupling in various watersheds. As discussed, the primary inputs are source unmixing, pathway decomposition,

and geospatial connectivity, all of which can be calculated utilizing a wide range of tools, for example, with loadograph separation, stable isotope tracing, and connectivity indices.

Conflict of Interest

We have no conflicts of interest to report.

Data Availability Statement

Calibration data, computer code, and model results will be stored and publicly available on Open Science Framework (at <http://www.doi.org/10.17605/OSF.IO/YKD86>).

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